Asymmetric Factor Volatility and Tail Risk: GARCH vs EGARCH in U.S. Equity Factors

Evidence from U.S. equity factors

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Agenda

1. **A-1. Motivation & stakes:** why factor volatility misspecification matters
2. **B-2. Data & design:** factors, regimes, models, evaluation
3. **D-3. Core results:** asymmetry, persistence, VaR backtests, regimes
4. **C-4. Portfolio implications:** correlation convergence & conditional frontier
5. **E-5. Conclusion:** what changes for PMs / risk, limitations, extensions

A-1 | The risk problem: why factor volatility modeling matters

- Risk systems rely on volatility forecasts for **position sizing, VaR, margin, hedging timing**.
- Standard symmetric GARCH assumes: *positive and negative shocks of same magnitude* $\Rightarrow$ *same vol response*.
  - **Non-normal returns:** fat tails + negative skew across factors
  - **Asymmetric volatility transmission:** "bad news" increases volatility more than "good news"

Practical consequence

Underestimating volatility in crisis peaks can delay hedging and understate tail risk precisely when correlation rises and diversification collapses.

A-1 | Executive summary

1. **Asymmetry is universal:** EGARCH leverage $\gamma < 0$ for all factors (2014–2026).
2. **Tail risk calibration improves:** EGARCH reduces VaR exceedances vs GARCH (SPX example).
3. **Regimes matter:** risk-adjusted performance varies materially by volatility regime.
4. **Diversification is state-dependent:** correlations are structurally high; rise further in stress.
5. **Portfolio optimization shifts:** conditional covariance pushes min-var into defensive concentration.

B-2 | Data universe

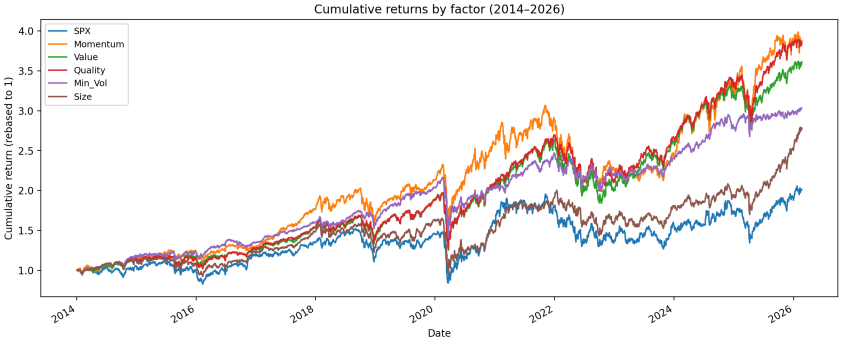
- Daily returns, 2014–2026
- **Market:** SPX proxy (SPY)
- **Style factors (ETFs as proxies):**
  - Momentum (MTUM)
  - Value (VLUE)
  - Quality (QUAL)
  - Minimum Volatility (USMV)
  - Size (IWM)
- Focus: conditional volatility, asymmetry, tail risk, and allocation implications.

Deliverable

A volatility-and-regime aware framework that tells a PM/risk team:

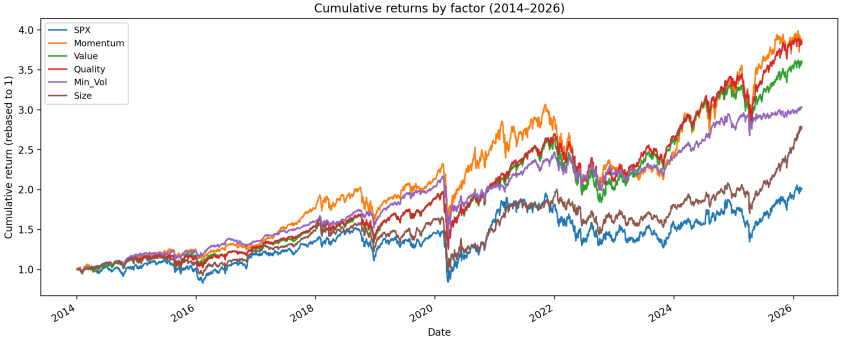
- when symmetric models break,
- how tail metrics react,
- how allocation set compresses.

B-2 | Visual context: factor performance over the full sample



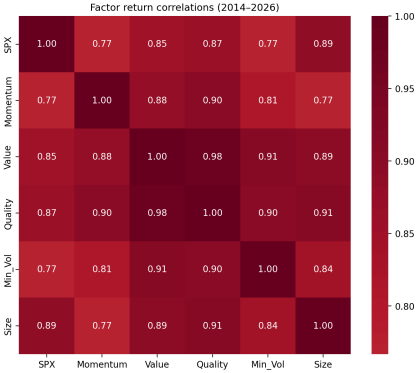
Key implication: Cumulative behavior differs, but risk is driven by *time-varying volatility* and *time-varying dependence*.

B-2 | Visual context: factor performance over the full sample



Key implication: Next: show dependence is *structurally high* (not orthogonal premia).

B-2 | Dependence: unconditional correlations are already high



- Correlations are uniformly high (roughly 0.77 to 0.98 in your heatmap).
- **Key implication:** Multi-factor diversification is **weaker than textbook intuition** even before conditioning on regimes.

B-2 | Model set (kept simple in main deck)

Symmetric baseline

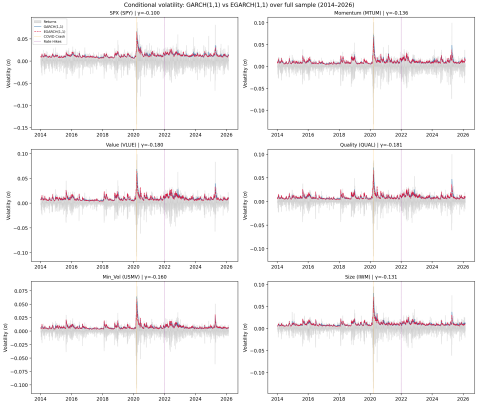
GARCH(1,1): volatility reacts to squared shocks symmetrically.

Asymmetric extension

EGARCH(1,1): log-variance + leverage term.

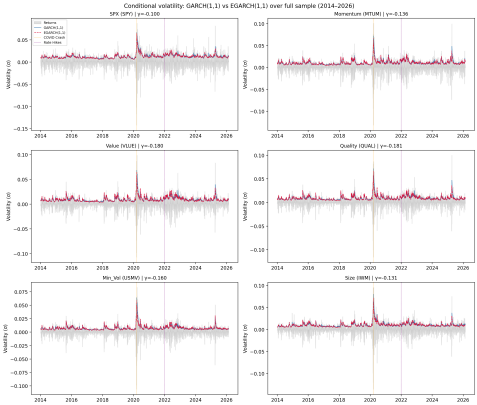
Key implication: The **sign** of shocks matters: negative returns can increase future volatility more.

D-3 | Conditional volatility: GARCH vs EGARCH (all factors)



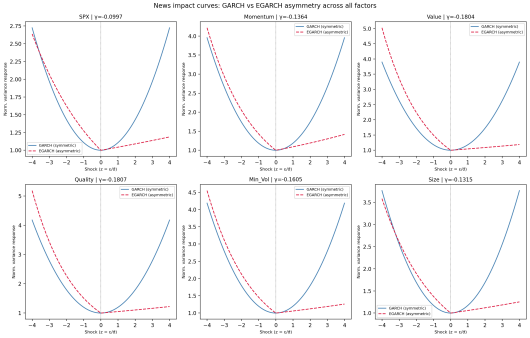
Key implication: Both models cluster volatility; differences are largest around crisis spikes.

D-3 | Conditional volatility: GARCH vs EGARCH (all factors)



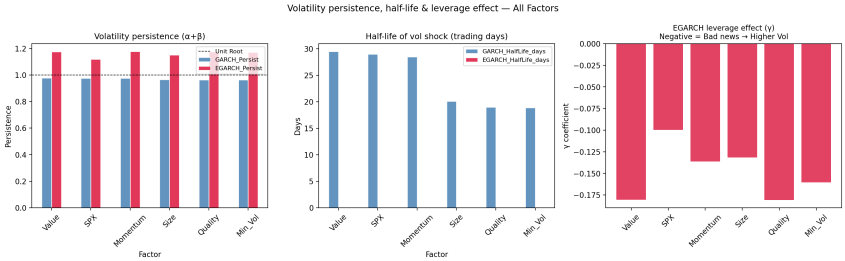
Key implication: EGARCH tracks downside-driven volatility surges more aggressively (consistent with leverage/asymmetry).

D-3 | Asymmetry: news impact curves (leverage effect)



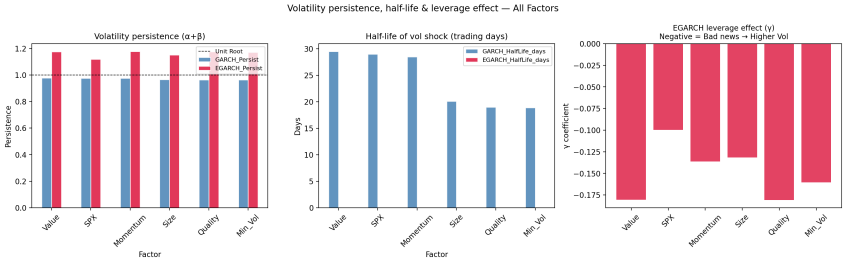
- In your estimates, $\gamma < 0$ **across all series** $\Rightarrow$ bad news increases volatility more than good news.
- **Key implication:** Even defensive sleeves (Quality, MinVol) show material asymmetry.

D-3 | Persistence, half-life, leverage (summary)



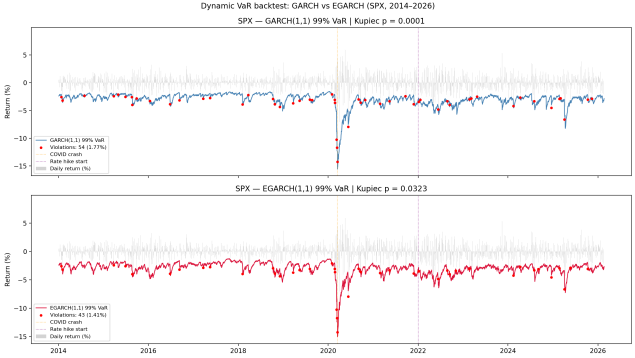
- GARCH persistence near unity implies shocks decay over **weeks**.
- **Key implication:** Regimes are sticky enough to matter for allocation and risk budgeting.

D-3 | Persistence, half-life, leverage (summary)



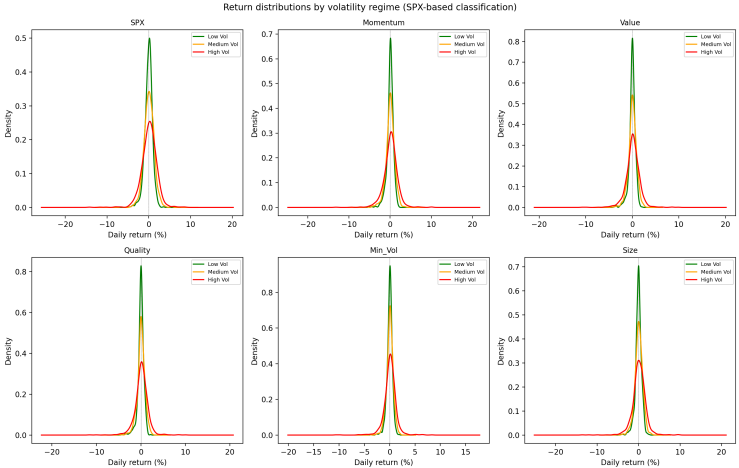
- **Leverage ranking:** Value/Quality strongest (largest negative γ), SPX weakest.
- **Key implication:** Style factors can amplify downside volatility transmission vs the broad market.

D-3 | Tail risk calibration: SPX 99% VaR backtest



- Visual: exceedances cluster in stress.
- **Key implication:** EGARCH reduces exceedances vs GARCH in SPX backtest, but extreme tails still challenge Gaussian GARCH-class models.

D-3 | Regime logic: volatility states & distributional shifts



Key implication: High-vol regimes show fatter tails across factors (distributional deterioration in stress).

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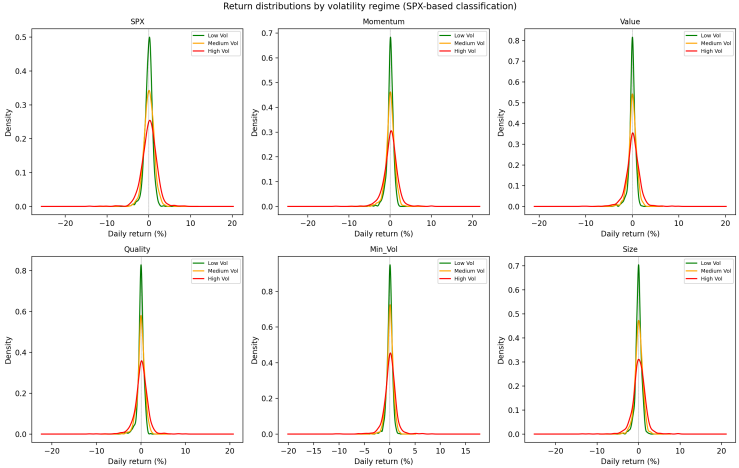
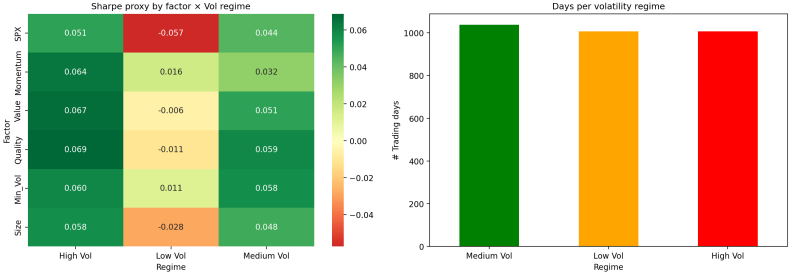


Key implication: This is where tail metrics and diversification assumptions break simultaneously.

D-3 | Regime counts + conditional Sharpe proxy (portfolio-relevant)



- **Risk-adjusted performance is state-dependent**.
- **Key implication:** Low-vol is not automatically the best regime for Sharpe-proxy across all sleeves.

C-4 | Mechanism: why diversification collapses in stress

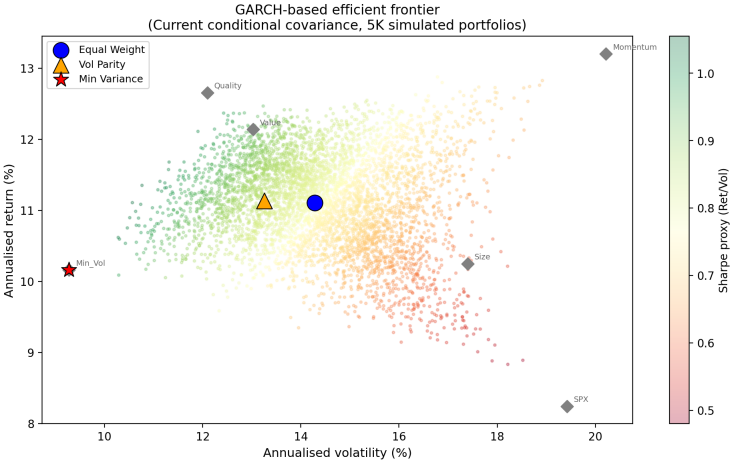
Three forces jointly

1. **Volatility persistence** (shocks last weeks)
2. **Leverage effect** ($\gamma < 0$; downside amplifies vol)
3. **Correlation convergence** (already high; rises in stress)

Portfolio-level outcome

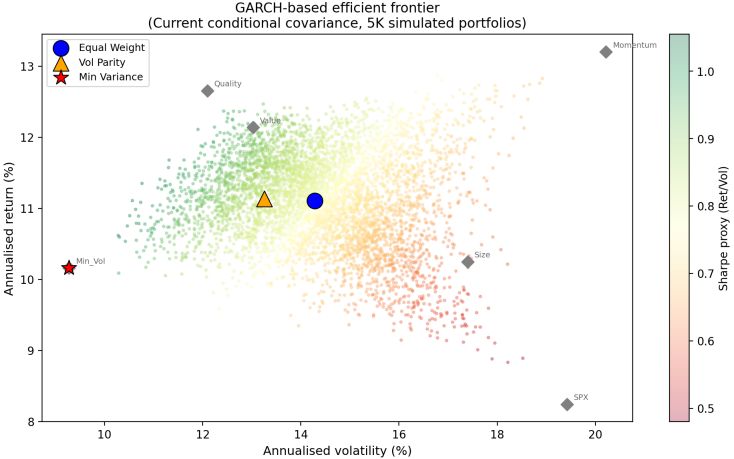
- Conditional covariance tightens
- Efficient frontier compresses
- Min-var can degenerate to defensive concentration

C-4 | Conditional frontier (GARCH-based covariance snapshot)



Key implication: Equal weight is not neutral under heteroskedasticity; it becomes an implicit risk bet.

C-4 | Conditional frontier (GARCH-based covariance snapshot)



Key implication: Min-variance concentrates in MinVol in your results: a live example of "diversification collapse" in a highly correlated factor set.

C-4 | What changes for PMs and Risk? (actionable)

Implementation-level guidance

- **Model choice should be regime-aware:** symmetric models can be fragile in downside shocks.
- **Tail calibration at 99% is the hard problem:** 95% can look fine while 99% fails.
- **Stress-time correlation is the hidden driver:** factor diversification can vanish when you need it.
- **Allocation should use conditional covariance:** volatility parity / defensive scaling can be a robust baseline vs naive min-var or equal weight.

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Key takeaway

This is not "EGARCH is always better." It is: **model asymmetry + regimes + dependence** $\Rightarrow$ **better risk calibration and allocation behavior when it matters**.

E-5 | Conclusions

1. **Universal leverage effect:** all factors show $\gamma < 0$ (bad news raises vol more).
2. **Tail risk is systematically underestimated in stress:** EGARCH improves but does not fully solve 99% tails.
3. **Regime dependence is economically meaningful:** volatility states persist long enough to guide risk budgeting.
4. **Diversification is state-dependent:** high structural correlations imply limited orthogonality among sleeves.
5. **Portfolio implication:** conditional covariance can force defensive concentration; risk parity is a pragmatic middle ground.

E-5 | Limitations

- Gaussian innovations: extreme tails may require **Student-t**, EVT overlays, or filtered historical simulation.
- Univariate modeling: dependence is treated largely via correlations and a conditional covariance snapshot; a next step is **multivariate DCC / asymmetric GARCH**.
- ETF proxies: factor implementation details (index methodology, rebalancing, transaction costs) can matter.

Key takeaway on limitations

Even with conservative modeling choices, the data still delivers the same core story: **asymmetry + persistence + high dependence $\Rightarrow$ fragile tail risk and compressed frontier in stress**.

E-5 | Extensions

Risk-model extensions

- EGARCH with Student-t innovations
- EVT overlay on standardized residuals
- CoVaR / systemic tail measures (multi-sleeve)

Allocation extensions

- Dynamic volatility targeting by sleeve
- Regime-conditioned covariance + stress overlays
- Robust optimization under correlation spikes

Appendix | Model equations

GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

EGARCH(1,1)

$$\ln(\sigma_t^2) = \omega + \beta \ln(\sigma_{t-1}^2) + \alpha \left(\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - \mathbb{E}|z| \right) + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}}$$

Key implication: Interpretation: $\gamma < 0$ means downside shocks drive disproportionately higher future volatility.

Appendix | Kupiec POF test (unconditional coverage)

Kupiec (1995) likelihood ratio

$$LR_{\mathrm{POF}} = -2\ln\left(\frac{(1-\alpha)^{N-x}\alpha^{x}}{(1-\hat{p})^{N-x}\hat{p}^{x}}\right)$$

- N = number of VaR forecasts, x = number of exceedances, α = nominal tail prob.
- Under correct coverage, $LR_{\mathrm{POF}} \sim \chi^2(1)$ asymptotically.

Appendix | References

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